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T11 – 05

Written Assignment 2

Aim: Introduction to Responsive Web Design and PWA

1. What is a Progressive Web Application (PWA)? Highlight some benefits of PWA.

A Progressive Web Application (PWA) is a type of application software delivered through the web, built using common web technologies including HTML, CSS, and JavaScript. PWAs are intended to work on any platform that uses a standards-compliant browser, including both desktop and mobile devices.

Benefits of PWA:

- Works offline or on low-quality networks.
- Can be installed on the user's device like a native app.
- Fast loading and responsive design.
- Push notifications support.
- Secure (served over HTTPS).
- Easy maintenance and updates.

2. Explain some requirements to make the website installable as PWA.

To make a website installable as a PWA, the following requirements must be met:

- A valid Web App Manifest file (manifest.json) with:
 - name or short_name
 - start_url
 - icons (192x192 and 512x512)
 - display (standalone or fullscreen)
- A service worker registered in the root scope to manage offline behavior.
- The website must be served over HTTPS.
- Pages must be responsive and work on all devices.
- The site should include a prompt or allow user-initiated install option.

3. Differentiate between PWA and Native App.

Feature	PWA (Progressive Web App)	Native App
Platform	Platform-independent	Platform-dependent (requires separate apps for iOS/Android)
Dependency	(runs in browser)	
Installation	Can be installed via browser (Add to Home Screen)	Installed from app stores like Google Play or App Store
Development Cost	Lower (single codebase for all platforms)	Higher (different codebases for each platform)
Performance	Slightly lower performance compared to native	High performance and access to device capabilities
Access to Device Features	Limited (depends on browser support)	Full access (camera, GPS, Bluetooth, etc.)
Offline Functionality	Supported via service workers	Fully supported
Updates	Automatic via browser	Requires app store approval and user action
Distribution	Via URL and can be shared easily	Through official app stores
User Experience (UI/UX)	Good, but depends on browser	Excellent, tailored to the platform
Storage Access	Limited	Full access